

SKILL WEEK – 2

TASK 1

```
nikith@NIKITH: ~/OSSP
Get:1 http://archive.ubuntu.com/ubuntu resolute/universe amd64 tree amd64 2.3.1-1 [53.5 kB]
Fetched 53.5 kB in 0s (274 kB/s)
Selecting previously unselected package tree.
(Reading database ... 40664 files and directories currently installed.)
Preparing to unpack .../tree_2.3.1-1_amd64.deb ...
Unpacking tree (2.3.1-1) ...
Setting up tree (2.3.1-1) ...
Processing triggers for man-db (2.13.1-1build1) ...
nikith@NIKITH:~/OSSP$ tree
.
├── LICENSE
├── Makefile
├── README.md
├── assets
│   └── architecture.png
├── bin
├── docs
│   ├── design.md
│   └── report.pdf
├── include
│   ├── builtin.h
│   ├── executor.h
│   ├── parser.h
│   └── shell.h
├── obj
├── scripts
│   └── run.sh
├── src
│   ├── builtin.c
│   ├── executor.c
│   ├── main.c
│   ├── parser.c
│   └── shell.c
└── tests
    ├── test_executor.c
    └── test_parser.c

9 directories, 18 files
nikith@NIKITH:~/OSSP$
```

TASK 2

```
#include <stdio.h>
```

```
#include <unistd.h>
```

```
#include <sys/types.h>
```

```
int main() {
```

```
    pid_t pid;
```

```
    pid = fork();
```

```
    if (pid < 0) {
```

```
        printf("Fork failed!\n");
```

```
    }
```

```
    else if (pid == 0) {
```

```
        printf("This is the Child Process.\n");
```

```

        printf("Child PID = %d\n", getpid());
    }

    else {

        printf("This is the Parent Process.\n");

        printf("Parent PID = %d\n", getpid());

        printf("Child PID = %d\n", pid);

    }

    return 0;

}

```

```

nikith@NIKITH:~$ man 2 fork
nikith@NIKITH:~$ man 3 exec
nikith@NIKITH:~$ nano fork_example.c
nikith@NIKITH:~$ gcc fork_example.c -o fork_example
nikith@NIKITH:~$ ./fork_example
This is the Parent Process.
Parent PID = 1039
Child PID = 1040
This is the Child Process.
Child PID = 1040
nikith@NIKITH:~$ nano fork_exec.c
nikith@NIKITH:~$ gcc fork_exec.c -o fork_exec
nikith@NIKITH:~$ ./fork_exec
Parent process.
Parent PID = 1113
Child PID = 1114
Child process created.
Child PID = 1114
total 152
drwxr-xr-x 10 nikith nikith 4096 Aug 24 05:02 OSSP
-rwxr-xr-x 1 nikith nikith 16480 Aug 17 05:16 command_executor
-rw-r--r-- 1 nikith nikith 1105 Aug 17 05:14 command_executor.c
-rwxr-xr-x 1 nikith nikith 16088 Aug 24 05:06 fork_example
-rw-r--r-- 1 nikith nikith 466 Aug 24 05:06 fork_example.c
-rwxr-xr-x 1 nikith nikith 16168 Aug 24 05:07 fork_exec
-rw-r--r-- 1 nikith nikith 630 Aug 24 05:07 fork_exec.c
-rwxr-xr-x 1 nikith nikith 15952 Aug 17 05:47 hello
-rw-r--r-- 1 nikith nikith 78 Aug 17 05:47 hello.c
-rwxr-xr-x 1 nikith nikith 16136 Aug 17 05:50 process_states
-rw-r--r-- 1 nikith nikith 433 Aug 17 05:50 process_states.c
-rw-r--r-- 1 nikith nikith 2091 Aug 17 05:06 task1.c
-rwxr-xr-x 1 nikith nikith 16248 Aug 17 05:32 task3
-rw-r--r-- 1 nikith nikith 1101 Aug 17 05:32 task3.c
-rwxr-xr-x 1 nikith nikith 16248 Aug 17 05:39 task4
-rw-r--r-- 1 nikith nikith 1302 Aug 17 05:38 task4.c
Child process completed.
nikith@NIKITH:~$ ls -l
total 152
drwxr-xr-x 10 nikith nikith 4096 Aug 24 05:02 OSSP
-rwxr-xr-x 1 nikith nikith 16480 Aug 17 05:16 command_executor

```